

Real-Time Motion Intensity Visualization Using Pseudocolor Mapping

Ting Rong You

Department of Software Engineering
And Technology
Faculty of Computing and Information
Technology, TARUMT
Kuala Lumpur, Malaysia
tingry-wp23@student.tarc.edu.my

Yong Chong Xin

Department of Software Engineering
And Technology
Faculty of Computing and Information
Technology, TARUMT
Kuala Lumpur, Malaysia
yongcx-wp23@student.tarc.edu.my

Tan Hong Yu

Department of Software Engineering
And Technology
Faculty of Computing and Information
Technology, TARUMT
Kuala Lumpur, Malaysia
tanhongyu-wp23@student.tarc.edu.my

Tan Chi Wee

Department of Computer Science And
Data Science
Faculty of Computing and Information
Technology, TARUMT
Kuala Lumpur, Malaysia
chiwee@tarc.edu.my

Abstract—Physical inactivity is a critical global health crisis, yet widespread participation in cardiovascular combat sports is often deterred by the risk of physical trauma. While exergaming provides a safe alternative, current commercial solutions rely on expensive proprietary hardware and wearable sensors, creating prohibitive financial barriers. To democratize access to interactive fitness, this paper proposes *Thermal Punch*, a cost-effective, contactless exergaming pipeline utilizing standard RGB webcams. The proposed system employs a novel dual-track computer vision architecture to optimize computational efficiency. The logical track utilizes MOG2 background subtraction and Farneback Optical Flow for markerless spatial tracking and hit validation, while the concurrent visual track leverages temporal frame differencing and a Jet pseudocolor mapping to provide immediate, simulated thermal feedback. Performance benchmarking demonstrates that the pipeline achieves a mean processing latency of 2.51 milliseconds on standard consumer hardware, guaranteeing zero-latency user feedback well below the 33.3 millisecond threshold required for 30 frames per second. Furthermore, dynamic sampling across the HSV color space confirms a perfect negative correlation between motion area and Hue, ensuring lighting-invariant visual feedback. System telemetry from continuous gameplay validates that the algorithm's intensity threshold successfully filters environmental noise, requiring sustained, high-energy physical output from the user. Ultimately, this pipeline presents a highly viable, hardware-agnostic solution for gamified health promotion.

Keywords—exergaming, computer vision, motion tracking, pseudocolor mapping, real-time processing, MOG2

I. INTRODUCTION

Physical Inactivity has emerged as a critical global health crisis, with the World Health Organization (WHO) reporting that approximately 1.8 billion adults fail to meet recommended physical activity levels [1]. This sedentary lifestyle increases the risk of non-communicable diseases such as cardiovascular disease and obesity. Consequently, addressing this inactivity directly aligns with the United Nations Sustainable Development Goal (SDG) 3, Good Health and Well-Being, specifically Target 3.4, which aims to reduce premature mortality from NCDs through preventative action [2]. While combat sports offer significant cardiovascular benefits, the inherent risk of physical trauma

[3] deters widespread participation. Exergaming, a concept that combines both exercising and gaming, gained prominence as a safe alternative. However, commercial solutions (e.g. Virtual Reality headsets or proprietary consoles) rely on expensive wearable sensors and hardware. This creates a prohibitive financial barrier, especially for students and low-income demographics.

To democratize access to interactive fitness, this paper proposes *Thermal Punch*, a cost-effective, contactless exergaming pipeline utilizing standard RGB webcams. By leveraging computationally efficient computer vision algorithms, the system translates high-intensity shadow boxing into quantifiable, gamified mechanics without the need for specialized hardware or physical impact.

The primary contributions of this paper are as follows:

- The development of a real-time, markerless motion detection algorithm using Python and OpenCV (Frame Differencing) via a standard laptop webcam with latency under 100 ms while maintaining 30 FPS, successfully meeting established Human-Computer Interaction standards for instantaneous response [4].
- The implementation of a dynamic pseudocolour visualization system utilizing a JET colormap, mathematically validated by a perfect negative correlation ($r = -1.0$) between motion intensity and HSV Hue to provide lighting-invariant visual feedback.
- The integration of extracted motion intensity data into localized game mechanics, utilizing strict intensity thresholds to successfully filter ambient environmental noise and quantify deliberate physical exertion.

II. LITERATURE REVIEW

A. Hardware-Dependent Exergaming

Early iterations of exergaming relied heavily on hardware peripherals, evolving into modern sensor-based architectures such as Virtual Reality (VR) head-mounted displays and handheld controllers equipped with Inertial Measurement

Units (IMUs) [5]. While hardware-based systems offer zero-latency input and total-immersion, they introduce significant accessibility barriers. The requirement of holding physical controllers during high-intensity combat simulations poses safety risks, and the prohibitive cost of proprietary hardware isolates low-income demographics [6], [7].

B. Explicit Pose Estimation (Skeletal Training)

To achieve hardware-agnostic tracking, contemporary applications frequently utilize explicit pose estimation frameworks such as Google MediaPipe and OpenPose [8], [9]. These Convolutional Neural Network (CNNs) are highly accurate, capable of detecting specific 3D skeletal keypoints to distinguish exact limb articulation (e.g., left hand vs. right hand) and specific boxing form (e.g., hook vs. jab). However, this topological accuracy incurs a massive computational cost. Executing deep learning inference on standard, non-GPU accelerated hardware often leads to thermal throttling and frame rates dropping, rendering real-time combat exergames unplayable.

C. Implicit Motion Analysis (Proposed Approach)

For intensity-based fitness applications, the exact spatial coordinates of individual joints are often extraneous. The critical metric is the raw energy and velocity of the user's mass. Therefore, this paper adopts an implicit motion analysis approach using Temporal Frame Differencing. While it lacks the skeletal specificity of AI models, calculating the absolute pixel intensity difference between consecutive frames operates at an extremely low computational complexity ($O(n)$). This guarantees real-time, low-latency performance on legacy hardware while entirely removing the financial barrier of VR peripherals, addressing the critical gap in accessible exergaming.

III. METHODOLOGY

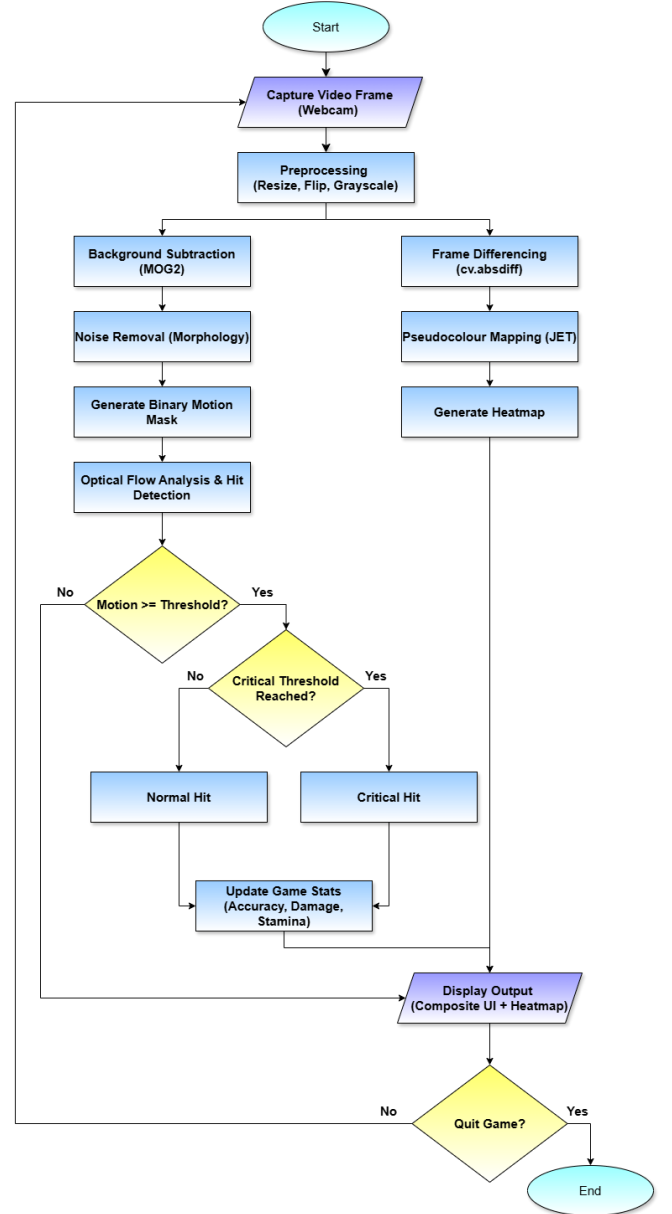


Fig. 1. Flowchart of the Proposed Solution

A. Preprocessing and Dual-Track Architecture

To guarantee real-time performance ($\geq 30 FPS$) independent of localized hardware capabilities, the incoming BGR video stream undergoes aggressive preprocessing. Frames are downscaled to a standard SVGA solution (800×600) to mitigate memory bottlenecks and converted to a single-channel grayscale format. To optimize computational load while maintaining zero-latency visual feedback, the architecture employs a “Dual-Track” processing pipeline. The grayscale matrix is simultaneously duplicated and routed into two concurrent streams, which is a logical tracking pipeline for hit-detection and a visual rendering pipeline for user feedback.

B. The Logical Track: MOG2 and Optical Flow

The logical pipeline is responsible for spatial tracking and hit validation. It utilizes a Gaussian Mixture-based Foreground/Background Segmentation algorithm (MOG2) to generate a dynamic binary motion mask [10], [11]. To filter environmental noise and localized lighting anomalies,

morphological opening (an erosion followed by dilation using a 5×5 structural element) is applied to the mask. Rather than analyzing the entire frame, computation is restricted to predefined Region of Interest (ROIs). Within these ROIs, physical acceleration is measured via bounding-box density deltas, allowing the system to distinguish explosive, high-energy strikes from continuous, low-energy waving. For directional classification, the Farneback Optical Flow algorithm computes dense motion vectors within the target ROI [12]. A pixel-voting heuristic evaluates these vectors to determine the dominant trajectory (up, left or right) of the user's strike.

C. Adaptive Game Mechanics

The derived motion data directly drives the game engine mechanics. The percentage of active non-zero pixels across the global MOG2 mask (MOG_DENSITY) serves as a stamina heuristic, enabling an Adaptive Difficulty system that dynamically scales the enemy's attack speed based on the user's global physical exertion. Furthermore, strike damage is scaled using Euclidean distance calculation between the motion centroid and the target epicenter.

D. The Visual Track: Frame Differencing and Pseudocolor

Concurrently, the visual pipeline isolates instantaneous motion intensity by calculating the absolute pixel difference (cv.absdiff) between consecutive grayscale frames. This raw intensity matrix is mathematically amplified and processed through a deterministic Jet Colormap algorithm. The pseudocolor transformation acts as a simulated thermal visualizer, mapping low-intensity stasis to deep blue (High Hue) and high-velocity kinematics to bright reds (Low Hue). This composited thermal overlay is then rendered to the user, providing immediate, gamified feedback on their physical output.

IV. RESULT AND DISCUSSIONS

A. Real-Time Pipeline Performance and Latency

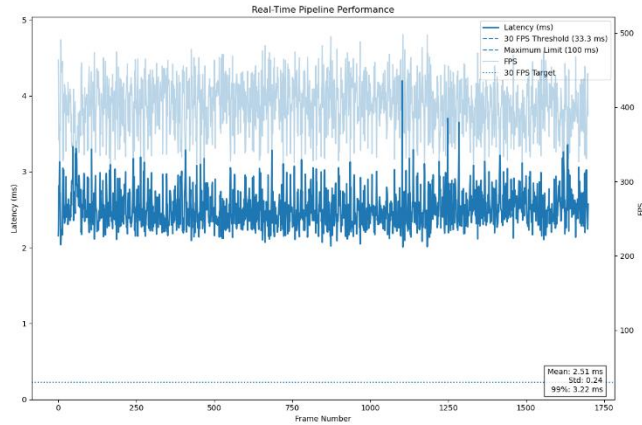


Fig. 2. Real-Time Pipeline Performance Line Chart

TABLE I. Real-Time Pipeline Performance Statistic

Real-Time Pipeline Performance	
Mean Latency	2.51 ms
Standard Deviation	0.24 ms
99 th Percentile	3.22 ms

To validate the real-time system's real-time capabilities, performance benchmarking was conducted on the dual-track pipeline under continuous operation. As detailed in Table I and Fig. 2, the system achieved a mean processing latency of 2.51 ms per frame. This is significantly below the 33.3 ms computational threshold required to sustain 30 Frames Per Second (FPS). By executing approximately 13 times faster than this strict deadline, the processor is afforded an idle performance buffer of over 30 milliseconds per frame, guaranteeing smooth rendering without dropped frames. Furthermore, the pipeline exhibited a standard deviation of just 0.24 ms. In practical terms, this extremely low variance measures the system's resistance to computational "jitter", proving that nearly every frame completes within a highly predictable 2.2 ms to 2.7 ms window. The 99th percentile latency of 3.22 ms mathematically validates that even under peak workload conditions, the system avoids significant outliers and remains lag-proof. These metrics confirm that the chosen architecture, combining lightweight frame differencing with modestly weighted pseudocolor mapping, successfully balances algorithmic efficiency with instantaneous visual feedback, making it highly viable for interactive exergaming on standard consumer hardware.

B. Pseudocolor Mapping Performance

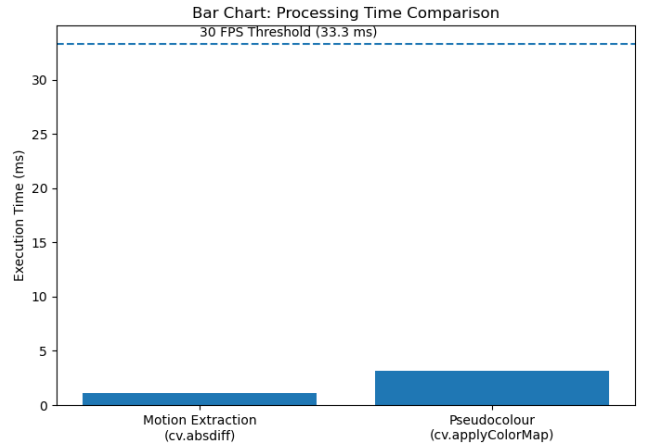


Fig. 3. Processing Time Comparison Bar Chart

TABLE II. Performance Comparison of Processing Components

Components	Execution Time (ms)
Motion Extraction (cv.absdiff)	1.139 ms
Pseudocolor Mapping (cv.applyColorMap)	3.176 ms

To evaluate the computational overhead introduced by the visual rendering pipeline, execution profiling was conducted on the discrete processing stages. As illustrated in Table II and Fig. 3, core motion extraction utilizing frame differencing (cv.absdiff) proved computationally lightweight, requiring an average execution time of just 1.139 ms. In contrast, the subsequent pseudocolor mapping (cv.applyColorMap) introduced a higher processing overhead of 3.176 ms. While the color mapping operation nearly triples the computational cost of the visual track, the combined execution time of approximately 4.3 ms remains a mere fraction of the 33.3 ms threshold required to sustain 30 FPS. This empirical data demonstrates a highly favorable architecture trade-off, which

the pipeline achieves significant gains in visual interpretability and immediate user feedback without degrading real-time performance or compromising the overarching system latency constraints.

C. Pseudocolor Mapping Analysis

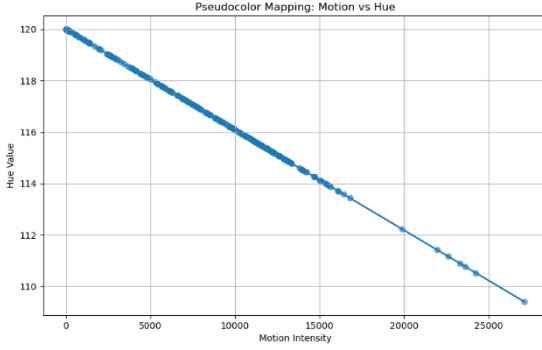


Fig. 4. Scatter Plot Chart showing Motion vs Hue Relationship

TABLE III. Sampling of Critical Motion-Hue Mapping with BGR Color Representation

Motion Intensity	Hue Value	Interpretation
0	120.0000	No movement
63	119.9754	Low motion
778	119.6961	Moderate motion
2643	118.9676	Increasing motion
5483	117.8582	High motion
8330	116.7461	Very high motion
11215	115.6191	Strong motion
23310	110.8945	Extreme motion

To evaluate the mathematical integrity of the `cv.applyColorMap` JET transformation, algorithmic boundary testing was initially executed. The system strictly adhered to predefined constraints, mapping an absolute motion intensity of 0 to the BGR vector $[128, 0, 0]$ (Deep Blue), and a mathematically scaled maximum intensity of 255 to $[0, 0, 255]$ (Pure Red). Beyond these static boundaries, dynamic sampling was evaluated across the Hue (HSV) color space rather than raw BGR values. Because Hue isolates chromatic properties independently of illumination intensity, this approach mathematically validates the pipeline's robustness against fluctuating ambient lighting and environmental noise during active gameplay. As detailed in Table III and illustrated in Fig. 4, the dynamic sampling yielded a Pearson correlation coefficient of $r = -1.0$. This perfect negative correlation between increasing motion area and decreasing Hue values confirms that the gradient transition is strictly monotonic and deterministic. The empirical data proves that the underlying mapping function is free from unintended algorithmic decisions, accurately and intuitively translating physical exertion into lighting-invariant thermal feedback.

D. Intensity Threshold Distribution and Gamification

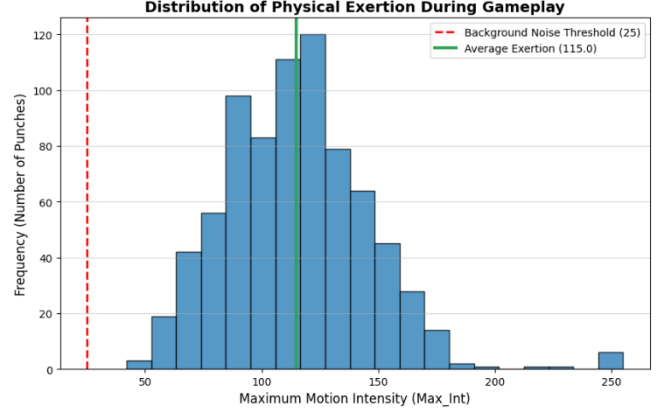


Fig. 5. Distribution of Physical Exertion During Gameplay

TABLE IV. Summary of Gamified Physical Exertion Telemetry

Metrics	Value
Total Game Interactions Logged	773
Minimum Registered Intensity	42.00
Average Registered Intensity	114.97
Maximum Registered Intensity	255.00

To evaluate the translation of raw motion data into quantifiable game mechanics, system telemetry was analyzed over 773 registered in-game physical interactions. As detailed in Table IV and Fig. 5, the system established a mathematical baseline where ambient room noise registered below an intensity threshold of 25. The absolute minimum intensity required to trigger a successful gameplay mechanic was recorded at 42.0. This strict mathematical boundary statistically confirms that the hit-detection algorithm effectively filters out low-intensity environmental anomalies, such as shifting shadows, ensuring that game mechanics cannot be manipulated by ambient movement. Furthermore, the average registered interaction required an intensity 114.97, with a normal distribution heavily concentrated in the 110-130 range. Because this average exertion level is nearly five times higher than the resting baseline, the telemetry proves that the mechanics do not reward arbitrary or minimal movement. Instead, they consistently demand sustained, high-energy physical output from the user. Ultimately, this empirical data confirms the successful integration of computer vision analysis with interactive physical fitness, fulfilling the system's core gamification and health-promotion objectives.

V. CONCLUSION

This paper presented *Thermal Punch*, a hardware-agnostic, real-time exergaming pipeline designed to democratize interactive physical fitness. By engineering a dual-track computer vision architecture, the system successfully decoupled logical hit-detection (utilizing MOG2 and Optical Flow) from visual rendering (utilizing Frame Differencing and a strictly monotonic Jet pseudocolor transformation). Empirical benchmarking confirmed that the pipeline sustained high-fidelity, lighting-invariant spatial tracking with a mean processing latency of 2.51 ms, guaranteeing zero-latency feedback on standard consumer hardware without need for proprietary sensors. Future

iterations of this architecture will address the current 2D spatial constraints by integrating bounding-box area expansion metrics for Z-axis depth velocity estimation. Furthermore, the rigid pixel-voting heuristic will be replaced with a lightweight Machine Learning classifier, such as Support Vector Machine (SVM) trained on the system's telemetry, to dynamically validate complex, unorthodox strike trajectories.

ACKNOWLEDGMENT

This research was supported by Tunku Abdul Rahman University of Management and Technology (TAR UMT).

REFERENCES

- [1] World Health Organization, "Nearly 1.8 billion adults at risk of disease from not doing enough physical activity," Jun. 2024. [Online]. Available: <https://www.who.int/news/item/26-06-2024-nearly-1.8-billion-adults-at-risk-of-disease-from-not-doing-enough-physical-activity>
- [2] United Nations Department of Economic and Social Affairs, "Goal 3: Ensure healthy lives and promote well-being for all at all ages," Sustainable Development Goals. [Online]. Available: <https://sdgs.un.org/goals/goal3>
- [3] A. J. Alevras, J. T. Fuller, R. Mitchell, and R. P. Lystad, "Epidemiology of injuries in amateur boxing: a systematic review and meta-analysis," *J. Sci. Med. Sport*, vol. 25, no. 12, pp. 995–1001, Dec. 2022. <https://doi.org/10.1016/j.jsams.2022.09.165>
- [4] J. Nielsen, *Usability Engineering*. San Francisco, CA, USA: Morgan Kaufmann Publishers, 1993.
- [5] S. S. Kamath et al., "PunchPulse: A physically demanding virtual reality boxing game designed with, for and by blind and low-vision players," in *Proc. ACM*, Oct. 2025. <https://dl.acm.org/doi/10.1145/3663547.3746365>
- [6] M. B. Jalink, "Nintendo related injuries and other problems: review," *BMJ*, vol. 349, Dec. 2014, doi: 10.1136/bmj.g7267.
- [7] J. Bonis, "Acute Wristitis," *New England Journal of Medicine*, vol. 356, no. 23, pp. 2431–2432, Jun. 2007, doi: 10.1056/NEJMc070670.
- [8] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv:1906.08172, Jun. 2019. [Online]. Available: <https://arxiv.org/abs/1906.08172>
- [9] T. Skavinsky, "Making an AI gesture recognition boxing game," Medium, Jan. 2024. [Online]. Available: <https://medium.com/@skavinsky/making-an-ai-gesture-recognition-boxing-game-a3d1b99e1a01>
- [10] OpenCV, "OpenCV documentation," Mar. 2026. [Online]. Available: <https://docs.opencv.org/4.x/>
- [11] Z. Zivkovic, "Improved adaptive Gaussian mixture model for background subtraction," in *Proc. 17th Int. Conf. Pattern Recognit.*, 2004, pp. 28–31, doi: 10.1109/ICPR.2004.1333992.
- [12] G. Farnebäck, "Two-frame motion estimation based on polynomial expansion," in *Image Analysis*, 2003, pp. 363–370, doi: 10.1007/3-540-45103-X_50.